LESSON 10.2

SOLVING ONE-STEP INEQUALITIES USING ADDITION AND SUBTRACTION

LESSON INTRODUCTION

MA.7.AR.2.1 Write and solve one-step inequalities in one variable within a mathematical context and represent solutions algebraically or graphically.

LEARNING TARGETS

- I can solve one-step inequalities in one variable using addition or subtraction.
- I can relate properties of inequality to solving one-step inequalities algebraically.

BENCHMARK COHERENCE

BUILDING ON

MA.6.AR.1.2 Translate a real-world written description into an algebraic inequality in the form of $x > a$, $x < a$, $x \geq a$, or $x \leq a$. Represent the inequality on a number line.

MA.6.AR.2.1 Given an equation or inequality and a specified set of integer values, determine which values make the equation or inequality true or false.

WORKING TOWARD

MA.8.AR.2.2 Solve two-step linear inequalities in one variable and represent solutions algebraically and graphically.

ESSENTIAL QUESTIONS

- How can I use properties of inequalities to solve one-step addition and subtraction inequalities?
- How do I show the solution of an inequality on a number line?

LESSON NARRATIVE

In this lesson, students first explore the relationship between inequalities and equations. Students also explore properties of inequality to get an understanding of the process of solving inequalities. Then students use properties of inequality to solve one-step addition and subtraction inequalities. Students also graph the solution sets and test values to prove their solution is true.

COMPONENT SUMMARY

10.2.1 WARM-UP *(~5 MIN.)*

Consider how cost estimators use inequalities

10.2.2 EXPLORATION *(10 MIN.)*

Explore properties of inequalities

10.2.3 GUIDED INSTRUCTION *(15 MIN.)*

Learn how to solve and graph one-step addition and subtraction inequalities

10.2.4 GUIDED PRACTICE *(10 MIN.)*

Practice solving and graphing one-step addition and subtraction inequalities

10.2.5 YOUR TURN! *(5-10 MIN.)*

Additional practice solving and graphing one-step addition and subtraction inequalities

10.2.6 WRAP-UP *(~5 MIN.)*

Compare and contrast the process for solving equations and solving inequalities

RESOURCES

GLOSSARY

Key Terms:

- Addition property of inequality
- Subtraction property of inequality
- Asymmetric property of inequality

Additional Terms:

- Inequality
- Boundary
- Solution set

MATERIALS

- Calculators
- (Optional) Number line templates
- (Optional) Colored pencils
- (Optional) Chart paper & markers

ADDITIONAL PREPARATION

- Consider preparing sentence stems or answer choices for constructive response problems.
- The 10.2.1 Warm-Up contains an optional video.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may struggle performing operations with rational numbers.
- Students may struggle to see how a>x is equivalent to x<a.
- Students may only recognize whole-number solutions and not the rational-number solutions.
- Students may incorrectly display the solution set on a number line for an inequality in the form of a>x.

THINGS TO CONSIDER

- Consider having no-calculator questions, such as the 10.2.2 Exploration, where students turn their calculator over. Teachers should look for and consider opportunities for students to improve their fluency with rational numbers.
- If time is an issue, then consider assigning 10.2.5 Your Turn! and/or 10.2.6 Wrap-Up for homework.

LITERACY AND LANGUAGE ROUTINES

Compare and Connect

In 10.2.3 Guided Instruction, students begin with solving an equation and then solve an inequality. This is a good point to ask students to compare and connect equations and inequalities. Ask them how the solving process and solution(s) are similar and different.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.2.1 Multiple Representations

Throughout the lesson, students represent inequalities algebraically and graphically. Specifically, in 10.2.5 Your Turn!, students solve inequalities and represent the solution set on a number line. These problems help students make connections among the multiple representations.

DIFFERENTIATE INSTRUCTION

For the 10.2.6 Wrap-Up, consider allowing students to circle the parts of the equation and inequality that are similar instead of writing an explanation.

10.2.1 WARM-UP: COST ESTIMATORS

ENRICHMENT

- What constraints on the costs might exist based on the project or industry?
- Can you think of any other careers that use inequalities?

10.2.1 Warm-Up: Cost Estimators

Cost estimators use data analytics to estimate the cost of manufacturing a product, constructing a building, or providing a service. They predict the amount of time, money, materials, and labor that will be needed to complete a project. This includes factoring in variables that may arise during the course of a project such as bad weather or shipping delays.

1. How do you think cost estimators use the minimum and maximum values of inequalities in their work?
   Answers will vary. Cost estimators use their price information to predict the most a service could cost and the least amount it could cost, this is like less than or equal to and greater than or equal to. Cost estimators factor in variables such as bad weather and give a high estimate on the most it may cost, but in reality it should be less than that amount.

2. Cost estimators usually specialize in a particular product or industry. Why do you think this is helpful?
   Answers will vary. It is helpful because they can become experts on how much a certain item or service should cost. It is helpful because they are knowledgeable about prices and possible situations that may raise the cost such as shipping delays and bad weather.

10.2.2 Exploration: Properties of Inequality

Three inequalities are shown.

$x > {}^{-}2.3$ $\quad$ ${}^{-}2.3 > x$ $\quad$ ${}^{-}2.3 < x$

1. Circle the inequalities that represent the same solution set.

2. Complete the statement.

   For both inequalities, all the values of x that are *greater* than -2.3 make both inequalities true.

These two inequalities are equivalent by the ***asymmetric property of inequality***.

The ***asymmetric property of inequality*** states that if $a > b$, then $b < a$.

3. Use the asymmetric property of inequality to write an inequality equivalent to $q + \frac{2}{5} \leq \frac{7}{10}$.

   $\frac{7}{10} \geq q + \frac{2}{5}$

4. Four true inequalities are given in the table. Work with your partner to complete the table by following the steps indicated for each inequality and then answering the questions.

Given Inequality	Step	What is the new inequality?	Is the new inequality true?
$-6 < -2$	Add 8.1 to both sides	$2.1 < 6.1$	*yes*
$-5 < \frac{3}{4}$	Subtract 2 from both sides	$-7 < -1\frac{1}{4}$	*yes*
$7.45 > 7$	Add $3\frac{1}{5}$ to both sides	$10.65 > 10\frac{1}{5}$	*yes*
$0 > -\frac{2}{3}$	Subtract $\frac{5}{6}$ from both sides	$-\frac{5}{6} > -1\frac{1}{2}$	*yes*

5. Use your work on the previous problem to complete the description of each property.

Property of Inequality	Description
Addition property of inequality	When the same value is added to both sides of a true inequality, … *the new inequality is also true.*
Subtraction property of inequality	When the same value is subtracted from both sides of a true inequality, … *the new inequality is also true.*

10.2.2 EXPLORATION: PROPERTIES OF INEQUALITY

DIFFERENTIATE INSTRUCTION

Consider having students practice reading inequalities aloud from right to left and left to right.

TEACHER MOVES

The Exploration is intentionally scaffolded to make connections between solving equations and inequalities. The Exploration also is scaffolded to introduce the properties of inequality that students apply when solving addition and subtraction inequalities. While circulating, support students who may need help.

TEACHER MOVES

Students should conclude that adding or subtracting the same value to both sides of an inequality results in a true statement. Provide a visual representation of why this is true by moving the two values in the inequality on the number line the same distance.

10.2.3 GUIDED INSTRUCTION: SOLVING ONE-STEP INEQUALITIES WITH ADDITION AND SUBTRACTION

TEACHER MOVES

10.2.3 Guided Instruction: Solving One-Step Inequalities with Addition and Subtraction

ENRICHMENT

Compare and Connect

- Highlight the connection between the equation and the inequality by considering the following questions.
- What changes when the equality symbol changes to an inequality symbol?
- How do the graphs of the solutions compare?

DIFFERENTIATE INSTRUCTION

Highlight the difference between the graph of the solution to an equation vs an inequality. Students do not often graph solutions to an equation on a number line. However, seeing the difference will provide a visual entry point.

QUESTIONING

Students may answer x>2 since that is the place where the value of x-3 first surpasses the number -2. Remind students that there are values between 1 and 2. Ask whether 1.1-3>-2.

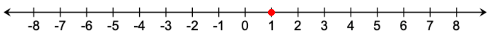
10.2.3 Guided Instruction: Solving One-Step Inequalities with Addition and Subtraction

1. Complete the table to determine the value of $x - 3$ for each value of x.

x	-4	-3.2	-2	$-\frac{1}{2}$	0	1	2	$3\frac{6}{7}$	4
$x - 3$	-7	-6.2	-5	$-3\frac{1}{2}$	-3	-2	-1	$\frac{6}{7}$	1

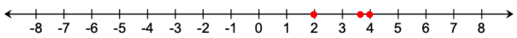
2. For which value of x is the equation $x - 3 = -2$ true?
   1

3. Represent the solution to the equation $x - 3 = -2$ on the number line.



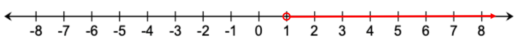
4. For which values of x in the table is the inequality $x - 3 > -2$ true?
   $2; 3\frac{6}{7}; 4$

5. Represent the values from the table that are part of the solution set of $x - 3 > -2$ on the number line.



The solution set of $x - 3 > -2$ includes more values than shown on the previous number line.

6. Graph the solution set for the inequality.



7. Write a value on the line to show an equivalent inequality represented by the graph above.

   $x > \underline{1}$

The solution set of the inequality $x - 3 > -2$ is all the values of x that are greater than 1, or $x > 1$.

This solution set can be determined algebraically by adding the same value to both sides of the inequality.

8. What value can be added to both sides of the inequality $x - 3 > -2$ to isolate x?
   3

Adding the same value to both sides of an inequality can be used to solve an inequality by the ***addition property of inequality***.

> ***Addition Property of Inequality***
>
> If $a > b$, then $a + c > b + c$.
>
> The ***addition property of inequality*** states that if the same value is added to both sides of an inequality, the result will be an equivalent inequality that is also true.

The ***subtraction property of inequality*** can also be used to solve an inequality.

> ***Subtraction Property of Inequality***
>
> If $a > b$, then $a - c > b - c$.
>
> The ***subtraction property of inequality*** states that if the same value is subtracted from both sides of an inequality, the result will be an *equivalent* inequality that is also *true*.

These properties can be used to solve one-step inequalities algebraically.

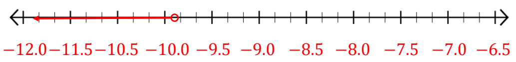
9. Use the inequality $-12.25 > y - 2.4$ to complete the following.

   a. Find the solution set by isolating the variable. Show your work.

   $$\begin{array}{rl} -12.25 & > y - 2.4 \\ \underline{+2.4} & \quad \underline{+2.4} \\ -9.85 & > y + 0 \\ -9.85 & > y \end{array}$$

   $y < -9.85$

   b. Graph the solution set on the number line.



   c. Use substitution to test a value or values in order to verify that the solution and the graph are correct.

   Answers will vary.

   $-12.25 > -10 - 2.4$
   $-12.25 > -12.4$

   $-12.25 > -12 - 2.4$
   $-12.25 > -14.4$

10.2.3 GUIDED INSTRUCTION: SOLVING ONE-STEP INEQUALITIES WITH ADDITION AND SUBTRACTION (CONTINUED)

DIFFERENTIATE INSTRUCTION

Students may benefit from first using concrete values instead of abstract variables. Consider having students substitute different integers for a,b, and c for a better understanding of addition and subtraction properties of inequality.

QUESTIONING

To expand students' thinking, have them discuss what other properties closely relate to these two properties of inequality. Students should be able to relate the following properties.

- The addition property of equality and the addition property of inequality.
- The subtraction property of equality and the subtraction property of inequality.

Prompt students to anticipate any other properties previously learned that might relate to inequalities.

TEACHER MOVES

Now that students have been formally introduced to the addition and subtraction properties of inequality, they should use them when solving inequalities. Ask students to compare the efficiency of algebraic methods compared to using substitution and Guess and Test.

QUESTIONING

Highlight the importance of checking answers by testing values from the graph in the inequality.

10.2.4 GUIDED PRACTICE: SOLVING ONE-STEP INEQUALITIES WITH ADDITION AND SUBTRACTION

DIFFERENTIATE INSTRUCTION

Students who struggle with operations involving rational numbers may need to be reminded to write the values in the same form (either all as decimals or all as fractions). If students struggle with these operations, consider allowing them to use a calculator. Have students write and say all computations that they enter in the calculator to reinforce their procedural fluency.

QUESTIONING

Question 2 highlights a common mistake. By correcting this mistake, students will be less likely to make it themselves.

TEACHER MOVES

Pause to discuss how the inequality represents the real-world situation.

ENRICHMENT

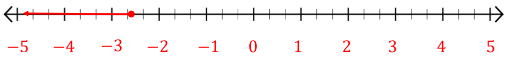
- How much would the repair bill be if the labor costs \$20? \$10? \$5?
- What is the least amount of money that could be charged for labor? Determine if it is possible that Michael's bill could be less than \$16.38. Explain your reasoning.

10.2.4 Guided Practice: Solving One-Step Inequalities with Addition and Subtraction

1. Use the inequality $-4\frac{3}{4}+c\le-7.4$ to complete the following.

   a. Solve the inequality. Show your work.
   $-4\frac{3}{4}+c\le-7.4$
   $4\frac{3}{4}-4\frac{3}{4}+c\le-7.4+4\frac{3}{4}$
   $c\ \le\ -2.65$

   b. Graph the solution set on the number line.



   c. Use substitution to test a value or values to verify that the solution and the graph are correct.
   Answers will vary.
   $-4\frac{3}{4}+-5\le-7.4$ $\quad$ $-4\frac{3}{4}+\ -3\le-7.4$
   $-9\frac{3}{4}\le-7.4$ $\quad$ $-7\frac{3}{4}\le-7.4$

2. Greg and Bianca both solved the inequality $-5.2+x>-8.4$. Their solutions are shown.

Greg's Solution	Bianca's Solution
$-3.2<x$	$x<-3.2$

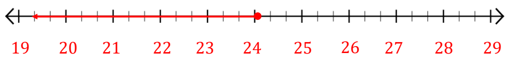
   Explain who is correct. Show your work.
   $-5.2+x>-8.4$
   $5.2-5.2+x>-8.4+5.2$
   $x>-3.2$
   Answers will vary. Greg is correct because you need to add 5.2 to both sides to isolate the variable. You then get $x>-3.2$. $x>-3.2$ is the same as $-3.2\ <x$. Bianca's solution is incorrect because x is greater than -3.2 not less than.

3. Michael got an estimate on the cost to repair his mountain bike. The parts will cost \$16.38 and he was told that the repair will be no more than \$40.50, including labor.

   The inequality $16.38+x\le40.50$ represents the situation where x is the cost of labor.

   a. Solve the inequality to determine the maximum amount the labor will cost.
   $x\le24.12$

   b. Graph the solution set on the number line below.



   c. Which solutions on the number line do not make sense for the context?
   Negative values do not make sense in this context because the amount charged for labor cannot be a negative value.

10.2.5 Your Turn!

Solve each inequality. Then, graph its solution set.

1. $y - 11 > -5$

   a. Solve the inequality and show your work.

   $y - 11 > -5$
   $\underline{+11 \quad +11}$
   $y > 6$

   b. Graph the solution set.

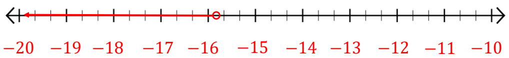
2. $-8\frac{1}{2} \geq x + 7.25$

   a. Solve the inequality and show your work.

   $-8\frac{1}{2} \geq x + 7.25$
   $-8\frac{1}{2} - 7.25 \geq x + 7.25 - 7.25$
   $-15.75 \geq x$

   $x \leq -15.75$

   b. Graph the solution set.



10.2.5 YOUR TURN!

DIFFERENTIATE INSTRUCTION

If students are struggling with which direction to shade on the graph, encourage them to substitute in a value from the graph to determine if it belongs in the solution set.

10.2.6 Wrap-Up: Cool Down

An equation and an inequality are shown.

$-2.25 = x + 2$ $\qquad$ $-2.25 \leq x + 2$

1. Describe how the processes of solving the equation and the inequality are similar.
   Answers will vary. The processes are similar because in both, you need to subtract 2 from both sides.

2. Complete the table to describe how the solutions to the equation and inequality are similar and different.

How are the solutions similar?	How are the solutions different?
• *Subtract 2 from both sides.* • *Both can be equal to.*	• *The inequality can be less than or equal to.* • *The equation has only one answer while the inequality has more than one answer.*

10.2.6 WRAP UP: COOL DOWN

DIFFERENTIATE INSTRUCTION

Consider allowing students to solve the equation and inequality and circle the parts that are similar rather than writing out an explanation.

